

The Five-Day Workweek and Spatial Labor Supply: Evidence from South Korea

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- ① Introduction
- ② Data
- ③ Empirical Analysis
- ④ Conclusion

Part I: Introduction

Motivation

- Labor supply adjusts not only through hours and participation, but also through **spatial margins** such as commuting and residential location
- Lower commuting frequency reduces the weekly burden of distance and reshapes residence–workplace matches
- Korea's Five-Day Workweek reduced commuting frequency from six to five days for covered workers
- But residential location is sticky: the relevant margin is likely to be workers whose residence–workplace match is still adjustable
- This creates a policy-induced change in the feasibility of living farther from Seoul jobs (e.g., Gyeonggi) among short-tenure workers

- **Main question:** How did Korea's Five-Day Workweek reshape the *Seoul job–Gyeonggi residence* margin?
 - ▶ Among short-tenure workers employed in Seoul, does statutory eligibility for the Five-Day Workweek increase Gyeonggi residence?
- **Mechanisms:**
Fewer commute days → lower shadow cost of distance → farther residence becomes feasible

This Paper

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 - ▶ Eligibility assigned by entry-firm size and sector

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 - ▶ Such workers employed in Seoul become about 5-6 percentage points more likely to reside in Gyeonggi

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 - ▶ Such workers employed in Seoul become about 5-6 percentage points more likely to reside in Gyeonggi
- Reframe working-time regulation as a determinant of **spatial labor-market adjustment**
 - ▶ Labor institutions can reshape where workers can feasibly live relative to their jobs

Related Literature

Commuting and Labor Supply

- Commuting time and distance affect labor supply, participation, and workplace choices
- Gutiérrez-i-Puigarnau and van Ommeren (2010); Black et al. (2014); Monte et al. (2018); Agarwal et al. (2018)

Five-Day Workweek Policy (Working time regulation)

- The Five-Day Workweek affected hours, employment, productivity, and well-being
- Ahn (2016); Ahn (2022); Lee and Lee (2016); Park and Park (2019)

This paper

- A working-time reform as a commuting-frequency shock to study residence–workplace re-optimization

Institutional Background: Five-Day Workweek in South Korea

- Pre-reform: Standard workweek was 44 hours, typically spread over six days (Saturday as a half-day)
- Legal basis: 2003 amendment to the Labor Standards Act introduced a **40-hour, five-day workweek**
- Policy objectives:
 - ▶ Improve work–life balance and working conditions
 - ▶ Harmonize labor standards with other OECD countries
 - ▶ Stimulate domestic consumption and service-sector activity
- Implementation: Reform was **staggered** across sectors and firm sizes, generating substantial variation in the timing of treatment

Institutional Background: Five-Day Workweek in South Korea

Phase	Effective date	Coverage (examples)
1	Jul 2004	Central government, public schools, banks; firms with $\geq 1,000$ employees
2	Jul 2005	Firms with 300–999 employees
3	Jul 2006	Firms with 100–299 employees
4	Jul 2007	Firms with 50–99 employees
5	Jul 2008	Firms with 20–49 employees
6	Jul 2011	Firms with 5–19 employees (near-universal coverage)

Staggered adoption of the statutory Five-day Workweek by firm size

Part II: Data

Data and Sample

- **Data:** Korea Labor and Income Panel Study (KLIPS), annual worker panel
- **Sample:** wage workers whose workplace is located in Seoul
- **Outcome:** probability of living in Gyeonggi among workers whose **workplace is Seoul**
- **Main analysis sample:** short-tenure workers with tenure 0–2
 - ▶ Early in a job match, residence–workplace arrangements are less settled and more likely to respond to commuting-cost changes

Empirical margin: Seoul workplace–Gyeonggi residence matches among short-tenure workers

Summary statistics (Seoul workplace)

	Full sample		Short-tenure workers		Difference
	Mean	SD	Mean	SD	
Age (years)	39.332	12.132	35.352	12.132	-3.980
Married (share)	0.558	0.497	0.387	0.487	-0.171
Log wage	5.108	0.678	4.865	0.633	-0.243
Years of education	5.972	1.345	5.973	1.275	0.001
Male (share)	0.566	0.496	0.477	0.500	-0.089
Weekly avg. working days	4.930	1.340	4.913	1.406	-0.017
Observations	14,250		6,178		

- Short-tenure workers are younger and less likely to be married, suggesting weaker residential attachment

Part III: Empirical Analysis

Treatment Assignment: Entry-based Statutory Eligibility

- The reform was rolled out by establishment size and sector
- I assign each worker i an **eligibility year** variable using entry-firm characteristics and sectors:

$$E_i \in \{2004, 2005, 2006, 2007, 2008, 2011, 0\}$$

- Treatment is defined by legal timing:

$$D_{it} = 1\{E_i > 0\} \times 1\{t \geq E_i\}$$

- E_i is fixed at entry: later job changes do not update treatment status
- This identifies an **Effect of statutory eligibility**

① Firm-size-based rollout

- Firm size may be correlated with job stability, industry composition, and residential patterns
- Addressed with industry $\times$ year FE and firm-size-specific linear trends

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- Endogenous job mobility or residential sorting may bias treatment comparisons
- Use short-tenure workers, fixed entry-based eligibility cohorts, and individual FE

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③ Concurrent metropolitan shocks

- Metropolitan-wide housing or transportation shocks may affect Gyeonggi residence
- Implement placebo tests using fake reform years, small firms, and non-Seoul workplace samples

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Identification Assumption

Conditional on fixed effects and controls, assigned eligibility timing is not systematically correlated with unobserved changes in Gyeonggi residence

Baseline Specification

$$Y_{it} = \beta D_{it} + X'_{it}\gamma + \alpha_i + \delta_{j(i,t),t} + \phi_{s(i,t)}t + \lambda_{o(i,t)} + \varepsilon_{it} \quad (1)$$

- $Y_{it} = 1\{\text{reside in Gyeonggi}\}$ among Seoul workplace worker i at time t
- β measures the change in the probability of Gyeonggi residence when assigned eligibility turns from 0 to 1
- α_i : individual FE; $\delta_{j,t}$: industry $\times$ year FE
- ϕ_{st} : firm-size-specific trends; λ_o : occupation FE
- E_i is assigned from entry-firm characteristics and fixed over time

Baseline: Short-Tenure Workers' Residential Adjustment

	Year + Industry FE	Year $\times$ Industry FE
	(1)	(2)
Post: $t \geq$ eligibility year	0.032* (0.016)	0.052** (0.024)
Observations	5,380	5,380
R^2	0.895	0.943

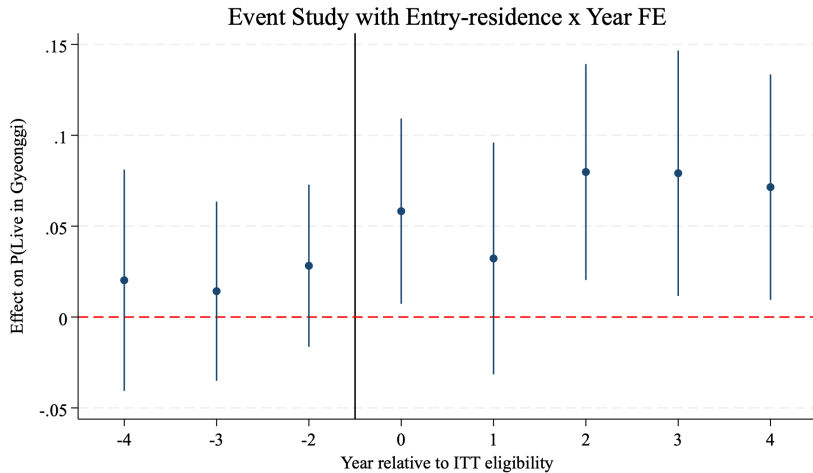
Column (1) includes year and industry FE; Column (2) includes industry $\times$ year FE. Both Specifications include individual FE, occupation FE, worker controls and firm-size trends. Standard errors are clustered at the individual level. ***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$.

Event-Study Specification

$$Y_{it} = \sum_{k \neq -1} \beta_k 1\{t - E_i = k\} + X'_{it}\gamma + \alpha_i + \delta_{j(i,t),t} + \phi_{s(i,t)}t + \lambda_{o(i,t)} + \varepsilon_{it} \quad (2)$$

- Same sample, controls, fixed effects, and trends as in the baseline specification
- Replaces the post-eligibility indicator with event-time indicators relative to E_i
- β_k measures the change in probability of Gyeonggi residence at event time k relative to one year before eligibility
- Lead coefficients test for differential pre-trends before statutory eligibility
- Lag coefficients trace the post-eligibility residential response

Event-Study: Short-Tenure Workers' Residential Adjustment



Avg. lead: 0.014 (0.022), joint $p = 0.881$, Avg. lag: 0.065** (0.028), joint $p = 0.018$

Robustness Checks: Placebo Tests

Question: Do similar timing patterns appear where statutory eligibility should not matter?

① Pre-reform placebo

- ▶ Fake reform in 2003: 0.002 (0.009); Fake reform in 2002: -0.009 (0.012)
- ▶ Weaken the concerns about pre-existing trends or anticipation

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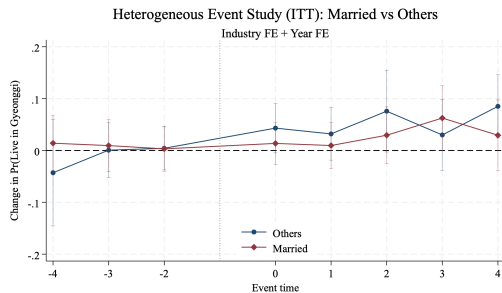
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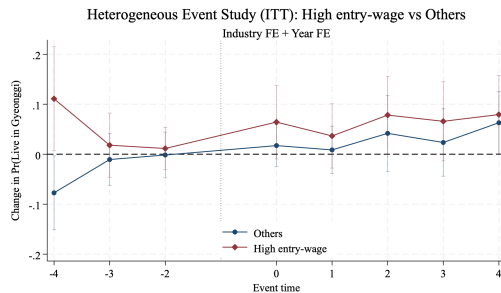
③ Non-Seoul workplace placebo

- ▶ Post reform: 0.002 (0.012); Event-study (Avg.lag): -0.007 (0.016)
- ▶ Less likely to be driven by metropolitan-wide shocks

Question: Who respond more to the reform?

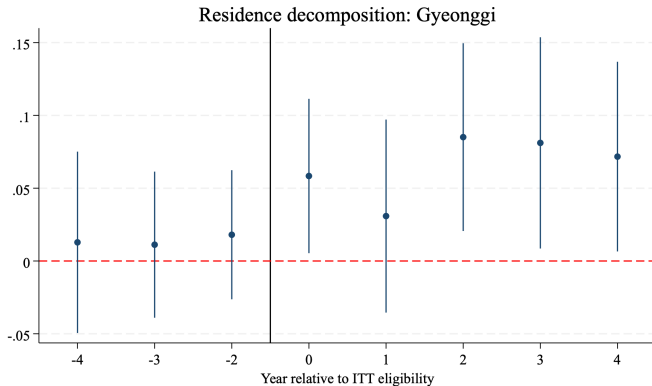


(a) *Heterogeneity by marital status*



(b) *Heterogeneity by wage (entry)*

Residence Decomposition: Gyeonggi

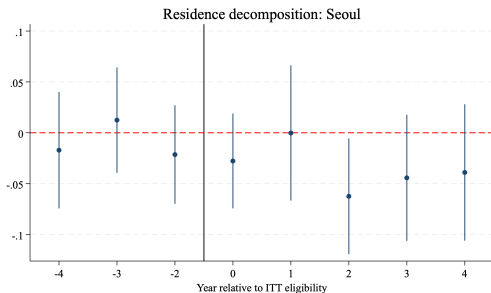


(a) Workers who live in Gyeonggi (Baseline Results)

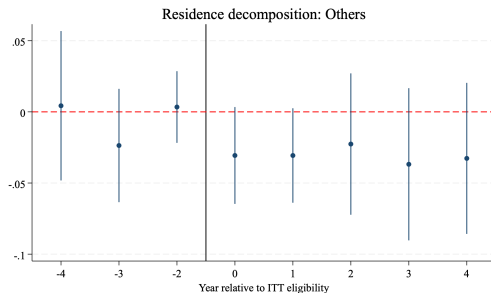
- Eligibility increases the probability of residing in Gyeonggi among Seoul workplace workers
- Next, I compare this with Seoul and other-region residence to locate the margin of adjustment

Residence Decomposition: Locating the Adjustment Margin

Question: Is this a Seoul–Gyeonggi adjustment or a broad non-Seoul shift?



(b) Workers who live in Seoul



(c) Workers who live in other regions

Part IV: Conclusion

- The Five-Day Workweek reform reduced commuting intensity by shifting covered workers from six to five workdays
- Using entry-based statutory eligibility, short-tenure workers employed in Seoul become more likely to reside in Gyeonggi after eligibility begins
- The response is concentrated on the Seoul–Gyeonggi residential margin

Working-time regulation can reshape the spatial organization of labor by changing commuting intensity

Thank You

Appendix

Appendix: Baseline Comparability

	Never-treated	Ever-treated	Difference
Age	37.770	35.270	-2.500
Education	5.221	5.814	0.593
Married	0.463	0.407	-0.057
Log wage	4.553	4.739	0.186

- Groups are not unconditionally balanced
- Identification relies on statutory timing conditional on FE, trends, and controls

Appendix: Assigned Eligibility Cohorts in the Sample

Eligibility year E_i	Workers	Share (%)
Never / < 5 workers	244	10.3
2004	1,566	66.3
2005	23	1.0
2006	47	2.0
2007	37	1.6
2008	107	4.5
2011	338	14.3
Total	2,362	100.0

- One observation per worker in the short-tenure Seoul-workplace analysis sample
- Eligibility year is assigned from entry-firm size and sector
- The distribution is concentrated in the first rollout cohort, motivating event-study validation and placebo tests

Appendix: Placebo Tests

	Fake reform	Small firms	Non-Seoul workplace
Estimate	0.002 / -0.009	-0.003	0.002
Avg. lag			-0.007

- Placebo estimates are small and statistically insignificant
- No comparable response appears before the rollout or outside the Seoul-workplace margin

Appendix: Robustness to Alternative Specifications

	Baseline	Window [-3,+3]	Cohort trends	Entry-res. $\times$ year FE
Avg. lag	0.065**	0.063**	0.049*	0.064**
SE	(0.028)	(0.028)	(0.032)	(0.027)
Lead p-value	0.881	0.729	0.687	0.634
Lag p-value	0.018	0.015	0.064	0.021

- Average post effects remain positive across alternative specifications
- Lead p-values remain large, providing no clear evidence of differential pre-trends

Appendix: Heterogeneity by Entry Residence

	Others	Seoul resident at entry
Avg. lead: $t = -4$ to -2	0.018	0.014
SE	(0.054)	(0.023)
Avg. lag: $t = 0$ to $+4$	0.208***	0.047*
SE	(0.078)	(0.026)
Diff. in avg. lag		-0.161**
p -value: diff.		0.031

- The post-eligibility response is significantly larger among workers already outside Seoul at job entry
- Difference in average post response is statistically meaningful (p -value: 0.031)

Appendix: Residence Decomposition

	Gyeonggi	Seoul	Other regions
Post	0.052**	-0.027	-0.024
SE	(0.024)	(0.022)	(0.015)
Avg. lead	0.014	-0.009	-0.005
SE	(0.022)	(0.021)	(0.015)
Avg. lag	0.065**	-0.035	-0.031
SE	(0.028)	(0.025)	(0.021)
<i>p</i> -value: lags	0.018	0.043	0.432

- The response is concentrated on the Seoul–Gyeonggi margin, not a broad shift to other regions